

SS-6: Deuteron Observables Beyond Binding: Scope and Limits of the Base-to-Base Picture

600-Cell Standard Model Emergence Series

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Thomas Lee Abshier, ND

Claude Opus (Anthropic)

Hyperphysics Institute, Kalispell, Montana

<https://hyperphysics.com>

drthomas007@protonmail.com

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Abstract

SS-5 derived the deuteron binding energy $B_d = 2.342$ MeV at leading order (+5.3%) from base-to-base bonding of hybrid-tetrahedral nucleons, along with concurrent predictions for ^3H , ^3He , and ^4He within 1.4%. A natural next question is whether the same bipyramid geometry predicts additional deuteron observables — the quadrupole moment Q_d , D-state admixture P_D , magnetic moment μ_d , matter radius r_d , and low-energy scattering parameters (a_{np} , r_0) — with comparable zero-parameter economy.

This paper reports a scoping exploration with one definitive negative finding and one positive direction. The negative finding: the rigid bipyramid's *intrinsic* quadrupole moment has the wrong sign — the three net- $+1/3$ electromagnetic charges produced by the base-to-base contact all sit in the equatorial plane, while the axial apices carry polarity (not EM charge), giving an oblate intrinsic $Q_d^{\text{int}} = -0.22$ fm² versus the observed prolate $Q_d = +0.286$ fm². This is not a failure of the mechanism; it reveals that Q_d , r_d , P_D , and μ_d are dominated by the *orbital* deuteron wavefunction at $r \sim 2$ fm, not by the bipyramid at ~ 0.4 fm. The positive direction: the zero-range Bethe-Peierls relation $a_{np} \approx 1/\kappa$ with $\kappa = \sqrt{2\mu B_d}/\hbar c$ gives $a_{np} = 4.32$ fm (−20% vs observed 5.43 fm), and the finite-range correction closes this gap in the correct direction but requires a derivation of the short-range potential $V_{\text{SR}}(r)$ from CPP primitives that has not yet been constructed.

We classify each deuteron observable into three categories: (A) bipyramid-geometric and hence computable at SS-5 precision (binding, spin-parity, isospin); (B) bipyramid-via- $V_{\text{SR}}(r)$ and hence computable once the short-range potential shape is derived (a_{np} , r_0 , virtual singlet); (C) orbital-dominated and hence requiring the full deuteron wavefunction beyond the bipyramid core (Q_d , r_d , P_D , μ_d). Two new open problems are registered: OPEN-SS-20 (derive $V_{\text{SR}}(r)$ from CPP primitives) and OPEN-SS-21 (derive the deuteron orbital wavefunction in the CPP framework). This paper does not attempt to close either.

Keywords: deuteron quadrupole moment, scattering length, effective range, D-state admixture, magnetic moment, intrinsic vs orbital contributions, short-range potential, Bethe-Peierls relation, zero-range approximation, bipyramid geometry, base-to-base nucleon configuration, 600-cell lattice, Conscious Point Physics, scoping analysis.

Plain Language Summary: SS-5 showed that gluing two tetrahedral nucleons together face-to-face correctly predicts their binding energy. A natural follow-up question is whether the same glued-tetrahedron geometry predicts *other* deuteron properties — its shape (quadrupole moment), its size (matter radius), and its scattering behaviour. This paper reports what we find when we ask this: the geometry predicts the binding cleanly, and gives a rough first-guess for how slowly-moving neutrons scatter off protons, but it cannot by itself predict the deuteron’s shape and size, because the deuteron spends most of its time with the two nucleons orbiting *around* each other at about 2 femtometres separation — much larger than the half-femtometre bipyramid core. So this paper is an honest scoping report: here is what the geometric picture predicts well, here is what it doesn’t, and here are the new pieces we would need to derive to get the rest.

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1 Introduction

SS-5 [1] derived four light-nuclei binding energies and three structural unboundness results from a single base-to-base bipyramid mechanism with zero fitted parameters. The question addressed here is whether the same mechanism predicts additional deuteron observables. If the bipyramid is a real geometric object and not a convenient mnemonic for the binding, then its charge distribution, spin structure, and coupling to external probes should produce definite predictions for Q_d , P_D , μ_d , r_d , a_{np} , and r_0 — all precisely measured.

This question is motivated by the “more stars for better navigation” principle: a mechanism that predicts one observable cleanly but fails on others is partially wrong; a mechanism that predicts several observables concurrently within the same residual band is likely structural. SS-5’s seven concurrent predictions (four quantitative bindings plus three unboundnesses) already constitute a strong test. Additional deuteron observables, if predicted at comparable precision from the same bipyramid, would further constrain the mechanism.

1.1 The finding, briefly

The result of this scoping is less triumphant than SS-5. Numerical exploration (§3) produces two clear outputs:

1. **The rigid bipyramid’s intrinsic quadrupole moment has the wrong sign.** The three net- $+1/3$ EM charges in the base-to-base contact plane give an *oblate* intrinsic charge distribution, $Q_d^{\text{int}} = -0.22 \text{ fm}^2$, whereas experiment shows *prolate* $Q_d = +0.286 \text{ fm}^2$. This is neither a failure of the mechanism nor a numerical precision issue: the signs are fundamentally opposite. Interpreted correctly, this result *reveals* that the observed Q_d cannot arise from the bipyramid core and must instead come from the *orbital* wavefunction at the deuteron’s $\sim 2 \text{ fm}$ scale — where the proton and neutron spend most of their time separated rather than in contact.
2. **The zero-range scattering length follows from B_d alone.** The Bethe-Peierls relation for a loosely-bound two-body state gives $a_{np} = 1/\kappa + r_0/2 + \dots$ with $\kappa = \sqrt{2\mu B_d}/\hbar c$. The leading (zero-range) term alone gives $a_{np} = 4.32 \text{ fm}$ with *no fitted parameters beyond the deuteron binding itself*, against the measured 5.43 fm (-20%). The gap closes in the right direction with a finite-range correction, but deriving r_0 requires knowing the shape of the short-range potential $V_{\text{SR}}(r)$, which CPP has not yet constructed from first principles.

The upshot is a classification: some deuteron observables are bipyramid-computable; some require an additional CPP input (the $V_{\text{SR}}(r)$ shape); some require the orbital wavefunction and are beyond the present reach.

1.2 What SS-6 delivers

- A definitive structural finding: the observed deuteron quadrupole moment is orbital-dominated, not bipyramid-dominated.
- A zero-range prediction $a_{np} = 1/\kappa = 4.32 \text{ fm}$ from B_d alone (-20% from measured), derived from the Bethe-Peierls relation without CPP-specific structure.
- A three-category classification of deuteron observables by derivability from the bipyramid.

- Two new open problems: OPEN-SS-20 ($V_{\text{SR}}(r)$ shape) and OPEN-SS-21 (deuteron orbital wavefunction).

1.3 What SS-6 does not deliver

- No prediction of Q_d , P_D , μ_d , or r_d from CPP — these are orbital-dominated.
- No derivation of $V_{\text{SR}}(r)$ shape from CPP primitives.
- No numerical computation of r_0 or the effective-range-corrected a_{np} .
- No improvement to SS-5's B_d prediction.

This is a scoping paper, not a result paper. Its purpose is to sharpen the programme's claims by distinguishing what the bipyramid mechanism actually predicts from what requires additional CPP input.

1.4 Open problems addressed

- **OPEN-SS-10** (Nuclear binding $V(r)$): partially clarified — the integrated binding at $A=2,3,4$ was resolved in SS-5; the short-range shape of $V_{\text{SR}}(r)$ is now separately registered as OPEN-SS-20.
- **New OPEN-SS-20**: derive the shape of $V_{\text{SR}}(r)$ as a function of inter-nucleon separation from CPP primitives. Controls a_{np} , r_0 , and the singlet virtual state.
- **New OPEN-SS-21**: derive the deuteron orbital wavefunction at ~ 2 fm from CPP in a framework that connects to the bipyramid core at ~ 0.4 fm. Controls Q_d , r_d , P_D , μ_d .
- **New PROP-SS-6-1**: the rigid bipyramid's intrinsic $Q_d^{\text{int}} = -0.22 \text{ fm}^2$ is oblate; observed Q_d is orbital-dominated.
- **New PROP-SS-6-2**: the zero-range Bethe-Peierls relation $a_{np} \approx 1/\kappa$ gives $a_{np} = 4.32$ fm from B_d alone (-20%).

2 Setup: The Bipyramid from SS-5

2.1 Geometric summary

In SS-5, the deuteron is the face-to-face contact of two hybrid-tetrahedral nucleons. Each nucleon (per SS-2) has:

- three quarks at the vertices of a triangular base face,
- one “open” vertex opposite the base carrying net polarity (+ for proton, – for neutron) but no additional electromagnetic charge,
- isosceles base geometry: u-u edge stretched to $r_{uu} = (1 + \varepsilon_{\text{cage}})l_{\text{edge}} = 1.07$ fm with $\varepsilon_{\text{cage}} = 1.94$; u-d edges at $r_{ud} = 0.62$ fm (SS-2).

In the base-to-base deuteron configuration, the proton base $\{u, u, d\}$ contacts the neutron base $\{d, d, u\}$ with the three charge-opposite vertex pairs aligned, forming a K_3 contact face. Three quark-quark DP chains span the contact, with K_3 collective mode reducing the three oscillators to one effective binding mode at $B_{\text{pair}} = M_0/\varphi = 2.342$ MeV (LO).

The open vertices of the two nucleons point *outward* along the symmetry axis: proton's + vertex at $z = +h$, neutron's - vertex at $z = -h$, with h the apex-to-base distance of a single cage. The resulting compound object is a *bipyramid*: two tetrahedral cages joined at a common triangular base, apices pointing outward.

2.2 Key geometric lengths

For the numerical scale of the bipyramid core, SS-2 gives:

$$\begin{aligned} l_{\text{edge}} &= 0.364 \text{ fm}, \\ l_{\text{unit}} &= \hbar c / \Lambda_{\text{QCD}} = 0.589 \text{ fm}, \\ r_{uu} &= 1.07 \text{ fm} \quad (\text{proton u-u edge}), \\ r_{ud} &= 0.62 \text{ fm} \quad (\text{u-d edges}), \\ h &\approx 0.63 \text{ fm} \quad (\text{apex-to-base; equivalent-regular-tet estimate}). \end{aligned}$$

The apex-to-apex long axis of the bipyramid is $2h \approx 1.26 \text{ fm}$. The equatorial extent is set by $r_{uu} = 1.07 \text{ fm}$. The bipyramid is mildly prolate, aspect ratio ~ 1.18 .

These scales are all $\lesssim 1.3 \text{ fm}$ — substantially smaller than the observed deuteron matter radius $r_d = 1.9753 \text{ fm}$ and charge radius $r_c = 2.1413 \text{ fm}$.

2.3 The separation of scales

A crucial physical fact: in the deuteron, the proton and neutron are *not* locked in the contact bipyramid geometry. They orbit at an rms separation of $\sim 4 \text{ fm}$ (inferred from $r_d^2 = \langle r_{np}^2 \rangle / 4$, giving $\langle r_{np}^2 \rangle^{1/2} \approx 3.95 \text{ fm}$). This is $\sim 10\times$ the bipyramid's internal scale and $\sim 4\times$ the bipyramid's apex-to-apex dimension.

This separation of scales is not a CPP issue — it is a standard feature of the deuteron recognized in all nuclear-physics treatments. The deuteron is a *loosely-bound* state: $B_d = 2.22 \text{ MeV}$, kinetic energy $\sim 20 \text{ MeV}$, potential depth $\sim 35 \text{ MeV}$ (standard values), giving a binding that is less than 10% of the characteristic nuclear energy scale. The wavefunction extends far outside the attractive-potential range. The asymptotic behaviour $\psi(r) \sim e^{-\kappa r} / r$ with $1/\kappa = 4.32 \text{ fm}$ sets the observed size.

This separation of scales is the dominant physics controlling Q_d , r_d , P_D , and μ_d . It is *not* controlled by the bipyramid core.

3 The Observables

We work through the non-binding deuteron observables in turn, identifying what the bipyramid can and cannot say.

3.1 Spin, isospin, parity

Already addressed in SS-5 §7 at the bipyramid level: $J^P = 1^+$, $I = 0$. These are geometric consequences of the base-to-base K_3 alignment and the S-wave relative motion. No additional work required. **Category A (bipyramid-geometric).**

3.2 Quadrupole moment Q_d

3.2.1 Intrinsic bipyramid quadrupole

Consider the bipyramid at rigid contact ($s = 0$ gap between the two bases). The three quark-quark DP chains span the contact with opposite-charge pairs superposed at identical (x, y) positions on the $z = 0$ contact plane:

- vertex 1: proton u ($+2/3$) superposed with neutron d ($-1/3$), net charge $+1/3$ at $(-r_{uu}/2, 0, 0)$;
- vertex 2: proton u ($+2/3$) superposed with neutron d ($-1/3$), net charge $+1/3$ at $(+r_{uu}/2, 0, 0)$;
- vertex 3: proton d ($-1/3$) superposed with neutron u ($+2/3$), net charge $+1/3$ at $(0, y_3, 0)$,

where $y_3 = \sqrt{r_{ud}^2 - (r_{uu}/2)^2} = 0.313$ fm is the base-triangle height from the u-u edge to the d vertex. Total net charge is $3 \times (+1/3) = +1$, the deuteron charge.

The apices at $z = \pm h$ carry polarity (one extra q CP each), not electromagnetic charge. They do not contribute to the intrinsic charge quadrupole.

For a charge distribution at positions $\vec{r}_i$ with charges q_i , the intrinsic quadrupole moment relative to the z -axis is

$$Q_d^{\text{int}} = \sum_i q_i (3z_i^2 - r_i^2).$$

With all three charges at $z_i = 0$, the formula reduces to

$$Q_d^{\text{int}} = - \sum_i q_i (x_i^2 + y_i^2) = -\frac{1}{3} \left[2 \cdot \frac{r_{uu}^2}{4} + y_3^2 \right] = -\frac{1}{3} \left[\frac{(1.07)^2}{2} + (0.313)^2 \right] = -0.224 \text{ fm}^2.$$

Finding 3.1 (Intrinsic bipyramid quadrupole has wrong sign). *The rigid base-to-base bipyramid, with electromagnetic charge concentrated in the equatorial contact plane and polarity-only apices along the axis, has intrinsic quadrupole moment $Q_d^{\text{int}} = -0.22 \text{ fm}^2$ (oblate). The measured deuteron quadrupole moment $Q_d = +0.286 \text{ fm}^2$ is prolate. The signs are opposite.*

3.2.2 Interpretation

This is not a failure of the bipyramid mechanism. It is a diagnostic. The observed Q_d is well-known in conventional nuclear physics to arise from the D-wave ($L = 2$) component of the deuteron wavefunction at orbital separation $r_{np} \sim 2$ fm. The relationship (standard):

$$Q_d \approx \frac{\sqrt{2}}{10} \int_0^\infty u(r)w(r)r^2 dr - \frac{1}{20} \int_0^\infty w(r)^2 r^2 dr,$$

where $u(r)$ is the S-wave radial wavefunction and $w(r)$ is the D-wave wavefunction, weights Q_d heavily with the large- r structure of the wavefunction. With $\langle r^2 \rangle \approx 15 \text{ fm}^2$ for the deuteron, the dominant contribution to Q_d comes from $r \gtrsim 2$ fm, far outside the $\lesssim 1$ fm bipyramid core.

The rigid bipyramid predicts -0.22 fm^2 ; the orbital wavefunction produces $+0.286 + 0.22 = +0.506 \text{ fm}^2$ of outward-pointing prolate contribution, so that the net observed value is $+0.286 \text{ fm}^2$. In other words, the orbital-dominated quadrupole contribution must be $\sim$ twice as large as the observed Q_d and of opposite sign to cancel the bipyramid's oblate intrinsic moment.

Proposition 3.2 (PROP-SS-6-1: Q_d is orbital-dominated). *The rigid base-to-base bipyramid produces an oblate intrinsic quadrupole $Q_d^{\text{int}} = -0.22 \text{ fm}^2$. The observed prolate $Q_d = +0.286 \text{ fm}^2$ therefore cannot arise from the bipyramid core alone; it is dominated by the orbital-scale ($r \gtrsim 2 \text{ fm}$) D-wave wavefunction. Quantitative prediction of Q_d from CPP therefore requires OPEN-SS-21 (orbital wavefunction), not bipyramid geometry alone.*

Category C (orbital-dominated).

3.3 Matter and charge radii r_d, r_c

$r_d = 1.975 \text{ fm}$ and $r_c = 2.141 \text{ fm}$ are both significantly larger than the bipyramid extent ($\lesssim 1.3 \text{ fm}$). In standard nuclear physics, $r_d^2 = \frac{1}{4}\langle r_{np}^2 \rangle + \frac{1}{2}(r_p^2 + r_n^2)$, with the first term (relative-motion) dominating.

The relative-motion contribution is controlled by the asymptotic form $\psi(r) \propto e^{-\kappa r}/r$ with $1/\kappa \approx 4.32 \text{ fm}$. The size of the deuteron is essentially the size of this exponential tail, not the bipyramid core. **Category C (orbital-dominated).**

3.4 D-state admixture P_D and magnetic moment μ_d

The D-state admixture is the fractional weight of the $L = 2$ component of the deuteron wavefunction. Modern NN-potential extractions give $P_D = 4.5\text{--}5.8\%$ (Argonne v18: 5.76%; CD-Bonn: 4.85%; chiral N³LO: 4.5–4.9%). The D-wave extends throughout the orbital region; its weight is fixed by the full wavefunction, not by the bipyramid’s intrinsic angular structure.

The magnetic moment μ_d follows from P_D via standard spin algebra:

$$\mu_d = \mu_p + \mu_n - \frac{3}{2}P_D(\mu_p + \mu_n - \frac{1}{2}),$$

with $\mu_p = 2.793\mu_N$, $\mu_n = -1.913\mu_N$. The experimental $\mu_d = 0.8574\mu_N$ implies $P_D \approx 3.9\%$ from this relation (slightly below the NN-potential extraction range, with the discrepancy attributed to meson-exchange and relativistic corrections).

Both P_D and μ_d are downstream consequences of the orbital wavefunction. **Category C.**

3.5 Scattering length a_{np} and effective range r_0

Low-energy np scattering in the triplet channel is described by the effective-range expansion

$$k \cot \delta_0 = -\frac{1}{a_{np}} + \frac{1}{2}r_0k^2 + \dots$$

For a loosely-bound two-body state, the Bethe-Peierls relation connects a_{np} to the bound-state wavenumber $\kappa = \sqrt{2\mu B_d}/\hbar c$:

$$\frac{1}{a_{np}} = \kappa - \frac{1}{2}r_0\kappa^2 + \dots,$$

so that

$$a_{np} = \frac{1}{\kappa} \left(1 + \frac{r_0\kappa}{2} + \dots \right).$$

With $B_d = 2.2246 \text{ MeV}$ and reduced mass $\mu = m_N/2 = 469.45 \text{ MeV}$:

$$\kappa = \frac{\sqrt{2\mu B_d}}{\hbar c} = 0.2316 \text{ fm}^{-1}, \quad \frac{1}{\kappa} = 4.318 \text{ fm}.$$

Proposition 3.3 (PROP-SS-6-2: Zero-range scattering length from B_d). *The zero-range Bethe-Peierls limit gives*

$$a_{np}^{(0)} = \frac{1}{\kappa} = 4.318 \text{ fm},$$

using B_d as the only input. The measured value is $a_{np} = 5.425 \text{ fm}$ [9], giving a leading-order error of -20.4% . The finite-range correction requires the effective range r_0 , which in turn requires the short-range potential $V_{\text{SR}}(r)$ (OPEN-SS-20).

3.5.1 Status

The 20% residual is outside the CPP $\sim 5\%$ generic band, so this is not a zero-parameter CPP prediction at SS-5 precision. However, it is a standard-physics result requiring only B_d as input. What CPP must supply, to go beyond zero-range precision, is $V_{\text{SR}}(r)$.

Inverting the Bethe-Peierls relation using measured a_{np} and κ :

$$r_0 \approx 2 \left(a_{np} - \frac{1}{\kappa} \right) = 2(5.425 - 4.318) = 2.21 \text{ fm}.$$

The experimental effective range is $r_0^{\text{exp}} = 1.749 \text{ fm}$ [10], so even this first-order inversion is $\sim 26\%$ off — the full expansion has significant higher-order terms for the deuteron.

Category B (bipyramid-via- V_{SR}).

3.6 Singlet virtual state

The 1S_0 np scattering channel has a virtual state near threshold: $E_v = +66 \text{ keV}$ above threshold (standard value). Analogously to the triplet, it is governed by the singlet channel $V_{\text{SR}}^{(s)}(r)$ which in base-to-base CPP corresponds to the K_3 face under orientation misalignment (SS-5 §?? discussion). Quantitative prediction again requires $V_{\text{SR}}^{(s)}(r)$. **Category B.**

4 Three-Category Classification

Table 1 summarizes the classification for all deuteron observables considered.

Table 1: Classification of deuteron observables by CPP derivability.

Observable	Measured	CPP category	Status
Binding energy B_d	2.2246 MeV	A (bipyramid)	Derived in SS-5 at $+5.3\%$
Spin $J^P = 1^+$	1^+	A (bipyramid)	Derived in SS-5
Isospin $I = 0$	0	A (bipyramid)	Derived in SS-5
Scattering length a_{np}	5.425 fm	B (via V_{SR})	Zero-range: 4.32 fm (-20%)
Effective range r_0	1.749 fm	B (via V_{SR})	Requires $V_{\text{SR}}(r)$; unknown
Singlet virtual state	+66 keV	B (via $V_{\text{SR}}^{(s)}$)	Requires singlet V_{SR}
Quadrupole Q_d	+0.286 fm ²	C (orbital)	Bipyramid intrinsic oblate, wrong sign
Matter radius r_d	1.975 fm	C (orbital)	Orbital-scale
Charge radius r_c	2.141 fm	C (orbital)	Orbital-scale
D-state admix P_D	4.5–5.8%	C (orbital)	Requires orbital WF
Magnetic moment μ_d	$0.857\mu_N$	C (orbital, via P_D)	Requires orbital WF

- **Category A (bipyramid-geometric, 3 observables):** binding, J^P , I . All derived in SS-5.
- **Category B (bipyramid-via- V_{SR} , 3 observables):** a_{np} , r_0 , singlet virtual state. Derivable once OPEN-SS-20 is solved.
- **Category C (orbital-dominated, 5 observables):** Q_d , r_d , r_c , P_D , μ_d . Derivable once OPEN-SS-21 is solved.

The classification is substantive: it tells a would-be critic that the bipyramid mechanism makes *no* claim about Q_d , r_d , P_D , or μ_d without additional CPP input. The Q_d -sign finding (Finding 3.1) *supports* this classification — the bipyramid actively predicts the wrong sign for Q_d if one were to apply it naively, which reveals that a different physical mechanism must dominate.

5 Open Problems

Open Problem 1 (OPEN-SS-20: Derive $V_{\text{SR}}(r)$ shape from CPP primitives). *Starting from the base-to-base K_3 face structure of SS-5, derive the shape of the short-range np potential $V_{\text{SR}}(r)$ as a function of inter-nucleon separation r . Constraints:*

- $V_{\text{SR}}(0) = -B_{\text{pair}} = -2.342 \text{ MeV}$ (LO, rigid contact).
- $V_{\text{SR}}(r) \rightarrow 0$ as $r \rightarrow \infty$.
- Range scale: $l_{\text{edge}} = 0.364 \text{ fm}$ (natural CPP scale).

Candidate functional forms: Coulombic $V_{\text{SR}}(r) \propto -l_{\text{edge}}/\sqrt{l_{\text{edge}}^2 + r^2}$; Yukawa-like $V_{\text{SR}}(r) \propto -\exp(-r/l_{\text{edge}})$; smooth cutoff $V_{\text{SR}}(r) \propto -l_{\text{edge}}/(l_{\text{edge}} + r)$. Each gives different numerical predictions for r_0 . The correct form must be selected by a CPP-structural argument, not a fit. Once derived, resolution gives:

- a_{np} beyond zero range,
- r_0 from first principles,
- singlet-channel $V_{\text{SR}}^{(s)}(r)$ and virtual-state energy,
- np phase shifts at low energy.

Open Problem 2 (OPEN-SS-21: Deuteron orbital wavefunction from CPP). *Derive the deuteron relative-motion wavefunction $\psi_{np}(r)$ in the CPP framework, connecting the bipyramid core at $r \lesssim 1 \text{ fm}$ to the orbital extension at $r \sim 2\text{--}4 \text{ fm}$. This is a multi-scale problem: at short range the K_3 face dynamics dominate; at long range the wavefunction is governed by the attractive potential tail plus the kinetic-energy balance appropriate for a loosely-bound state. Once resolved, yields:*

- r_d , r_c from $\langle r_{np}^2 \rangle$,
- P_D from $|w(r)|^2$ integrated weight,
- Q_d from standard quadrupole integrals including both intrinsic bipyramid contribution and orbital D-wave contribution,
- μ_d via the P_D -dependent magnetic-moment relation.

The central unresolved question is how the CPP K_3 collective mode extends from the rigid-contact limit to finite inter-nucleon separation in a way compatible with the observed orbital size.

6 Discussion

6.1 Why the negative finding is useful

The Q_d -sign result (Finding 3.1) is not a failure of CPP; it is a clarification. Three points:

- (i) It sharpens the mechanism’s claims. SS-5 establishes that the bipyramid predicts binding; SS-6 establishes that the bipyramid does *not* predict quadrupole. This is a falsifiable, testable distinction that any competing mechanism also has to navigate — if an alternative theory claims to predict both B_d and Q_d from the same rigid geometric object, that theory is suspect.
- (ii) It matches conventional nuclear physics. In the deuteron, Q_d is well-known to arise from the D-wave orbital tail, not from a short-range “shape.” CPP joining this understanding, rather than claiming to predict Q_d from the bipyramid, is consistent with known physics.
- (iii) It redirects programme effort. Instead of trying to squeeze Q_d , r_d , and P_D out of the bipyramid, the programme now has a sharply defined open problem (OPEN-SS-21) — derive the orbital wavefunction — which is the right target for the next serious theoretical effort.

6.2 The zero-range scattering length as a consistency check

The $a_{np} = 1/\kappa = 4.32$ fm zero-range prediction is not a CPP-specific result. Any theory that correctly predicts B_d automatically gets this zero-range prediction right at the same precision, via the Bethe-Peierls relation. What CPP must do is *go beyond* zero range by supplying $V_{SR}(r)$. The 20% zero-range residual is therefore a measure of how far CPP has to travel to claim a precision prediction for a_{np} .

6.3 Comparison with the SS-5 strategy

SS-5’s strength is a single mechanism giving seven concurrent independent predictions. SS-6’s contribution is a single mechanism giving *one* concurrent clarification: the bipyramid’s active prediction of an oblate intrinsic Q_d (wrong sign) reveals the orbital-dominance of the observed value. Compared to SS-5, this is a smaller paper with lower epistemic density, but it closes the easy avenue (“just predict more deuteron observables from the bipyramid”) cleanly and redirects effort toward the real open problems.

7 Conclusion

Applied to deuteron observables beyond the binding energy, the SS-5 base-to-base bipyramid mechanism yields three categories:

- **A (bipyramid-geometric):** binding, spin, isospin, parity — derived in SS-5.
- **B (bipyramid-via- $V_{SR}(r)$):** a_{np} , r_0 , singlet virtual state — await $V_{SR}(r)$ derivation (OPEN-SS-20).
- **C (orbital-dominated):** Q_d , r_d , P_D , μ_d — await orbital wavefunction (OPEN-SS-21).

The rigid-bipyramid intrinsic quadrupole has the wrong sign — a diagnostic result revealing that the observed Q_d is orbital-dominated. The zero-range scattering length from B_d alone is $a_{np} = 4.32$ fm (−20% from measured).

This paper does not close either of its two new open problems. Its contribution is scope clarification: it maps out which deuteron observables are downstream of the bipyramid (SS-5’s territory), which require one further CPP input (OPEN-SS-20), and which require a new CPP framework for the orbital regime (OPEN-SS-21). The programme gains two sharp open problems and one definitive negative finding in exchange for an honest assessment of what the base-to-base mechanism actually claims.

7.1 Problem status after this paper

- OPEN-SS-10 (nuclear $V(r)$): clarified — integrated binding at $A=2,3,4$ done in SS-5; short-range shape now separately OPEN-SS-20.
- NEW OPEN-SS-20 ($V_{SR}(r)$ shape): OPEN.
- NEW OPEN-SS-21 (orbital wavefunction): OPEN.
- NEW PROP-SS-6-1 (Q_d orbital-dominated, bipyramid intrinsic oblate): SUPPORTED by explicit calculation.
- NEW PROP-SS-6-2 (zero-range $a_{np} = 1/\kappa$ from B_d): DERIVED (standard Bethe-Peierls), –20% from experiment.

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